CHAPTER 5 · CONCEPT 2 · WORKSHEET

Divisibility & Digital Roots — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 · MCQ **[1]**

A number is divisible by 9 if:

(a) its last digit is 9 (b) the sum of its digits is divisible by 9 (c) its last digit is even
(d) it ends in 0

Q2 · MCQ **[1]**

To test divisibility by 11, you check whether:

(a) the last digit is 1 (b) the digit sum is a multiple of 11
(c) the alternating sum of digits is 0 or a multiple of 11 (d) the last two digits are divisible by 11

Q3 · FILL IN THE BLANK **[1]**

The digital root of 7309 is ______.

____

Q4 · MCQ **[1]**

The number 405 is divisible by:

(a) 2 (b) 4 (c) 9 (d) 11

Q5 · TRUE / FALSE **[1]**

A number divisible by 6 must be divisible by both 2 and 3.

True False

Q6 · ASSERTION–REASON [1]

Assertion (A): 1 is a prime number.
Reason (R): Prime numbers have exactly two factors.

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7 [2]

Without dividing, check whether (i) 405 and (ii) 8888 are divisible by 9.

Q8 [2]

Find the smallest and the greatest 4-digit number that is divisible by 9.

Q9 [2]

Find all values of z so that the number 31z5 is a multiple of 9.

Long answer (3 marks each)

Q10 [3]

Determine whether 406839 is divisible by 2, 3, 4, 5, 6, 9 and 11, using the divisibility tests.

Q11 **[3]**

Find the digital root of 63p + 18q + 29 for any integers p and q. Explain why it does not depend on p or q.

Q12 **[3]**

Find the smallest and the greatest 6-digit number that is divisible by 11.

Total: 21 marks · Check your work against the answer key.